Proposal for Highway Rest Area EV Charging Station DES

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1 Introduction

This proposal aims to design a Discrete Event Simulation (DES) model to analyze the efficiency of an EV charging station in a hypothetical highway rest area.
The model considers only vehicles entering for charging purposes and excludes movement time within the rest area. The chargers are limited to two types: fast chargers and slow chargers, assuming vehicles prefer fast chargers. The target vehicles are classified into three types: Truck, Sedan, and Compact Car. The charging time is determined solely by the charger type and the vehicle type.

2 System Model and Assumptions

This system applies the $M(t)/G/2/K$ model according to Kendall's Notation.

2.1 Arrival Process and Charging Time

Vehicle arrivals follow a Nonhomogeneous Poisson Process, where the arrival rate varies over time. The vehicle type entering the system is determined using the inverse transform method utilizing the cumulative distribution function (CDF) intervals of a Uniform RV. The charging time for slow chargers follows a general probability distribution (General RV), and the fast charger's service rate is set to three times that of the slow charger.

2.2 Servers and System Capacity

The charging station operates a total of 2 servers in parallel: 1 fast charger and 1 slow charger.
Reflecting the spatial constraints of the rest area, the maximum number of vehicles the system can accommodate is limited to K. That is, excluding the 2 vehicles currently charging, a maximum of $K-2$ vehicles can wait in the queue.

3 Simulation Model

3.1 Variable Definitions

- **Time Variable:** Defined as t.
- **System State Variable:** Represented as $SS = (n, i_1, i_2)$.
  - n : Total number of vehicles in the system ($0 \leq n \leq K$)
  - i_1, i_2 : Unique ID of the vehicle currently charging at the fast (No. 1) and slow (No. 2) chargers, respectively (0 if empty)
- **Counter Variables:**
  - N_A : Cumulative number of vehicles arrived up to time t
  - C_j : Number of vehicles that completed service at charger j up to time t ($j \in \{1, 2\}$)

- N_{loss} : Number of vehicles that left without entering the system due to capacity overflow

- **Output Variables:**
  - $A(n)$: Arrival time of the n-th vehicle ($n \geq 1$)
  - $D(n)$: Departure time of the n-th vehicle ($n \geq 1$)

3.2 Event List and Initialization

- **Event List:**
  - t_A : Scheduled arrival time of the next vehicle
  - t_1 : Scheduled service completion time for the fast charger (No. 1)
  - t_2 : Scheduled service completion time for the slow charger (No. 2)

- **Simulation Initialization:**
  - Variable initialization: $t = N_A = C_1 = C_2 = N_{\text{loss}} = 0$ and $SS = (0, 0, 0)$
  - First event scheduling: Generate the first arrival time random variable T_0 and set $t_A = T_0$
  - Server completion times: Set $t_1 = \infty$, $t_2 = \infty$ as they are not operating in the initial state

3.3 Recursive Loop

Case 1: New Customer Arrival ($t_A = \min(t_A, t_1, t_2)$)

- **Basic Variable Updates:**

- Time update: $t = t_A$
- Increment cumulative arrivals: $N_A = N_A + 1$
- Schedule next arrival: Generate a random variable T_t and set $t_A = t + T_t$
- Data collection: Record the arrival time of the current vehicle $A(N_A) = t$

- **System State (SS) Updates:**
  - **When capacity is exceeded (Balking):** If $n = K \rightarrow$ Reset $N_{\text{loss}} = N_{\text{loss}} + 1$
  - **When both are empty (No. 1 gets priority):** If $SS = (0, 0, 0) \rightarrow$ Reset $SS = (1, N_A, 0)$ and $t_1 = t + Y_1$
  - **When only No. 1 is in use:** If $SS = (1, j, 0) \rightarrow$ Reset $SS = (2, j, N_A)$ and $t_2 = t + Y_2$
  - **When only No. 2 is in use:** If $SS = (1, 0, j) \rightarrow$ Reset $SS = (2, N_A, j)$ and $t_1 = t + Y_1$
  - **When both are in use (Join queue):** If $2 \leq n < K \rightarrow$ Reset $SS = (n + 1, i_1, i_2)$

Case 2: Fast Charger (No. 1) Completion ($t_1 < t_A$ and $t_1 \leq t_2$)

- **Basic Variable Updates:**
  - Time update: $t = t_1$
  - Increment No. 1 cumulative processing: $C_1 = C_1 + 1$
  - Data collection: Record the departure time of the vehicle finishing service $D(i_1) = t$

- **System State (SS) Updates Based on Queue Status:**

- **If** $n = 1$**:** *(Only 1 vehicle was in the system)* $\rightarrow$ Reset $SS = (0, 0, 0)$ and $t_1 = \infty$
- **If** $n = 2$**:** *(No queue and only No. 2 is operating)* $\rightarrow$ Reset $SS = (1, 0, i_2)$ and $t_1 = \infty$
- **If** $n > 2$**:** *(Vehicles in queue)* $\rightarrow$ Calculate $m = \max(i_1, i_2)$, then set $SS = (n - 1, m + 1, i_2)$ and new completion time $t_1 = t + Y_1$

Case 3: Slow Charger (No. 2) Completion ($t_2 < t_A$ and $t_2 < t_1$)

- **Basic Variable Updates:**
  - Time update: $t = t_2$
  - Increment No. 2 cumulative processing: $C_2 = C_2 + 1$
  - Data collection: Record the departure time of the vehicle finishing service $D(i_2) = t$

- **System State (SS) Updates Based on Queue Status:**
  - **If** $n = 1$**:** *(Only 1 vehicle was in the system)* $\rightarrow$ Reset $SS = (0, 0, 0)$ and $t_2 = \infty$
  - **If** $n = 2$**:** *(No queue and only No. 1 is operating)* $\rightarrow$ Reset $SS = (1, i_1, 0)$ and $t_2 = \infty$
  - **If** $n > 2$**:** *(Vehicles in queue)* $\rightarrow$ Calculate $m = \max(i_1, i_2)$, then set $SS = (n - 1, i_1, m + 1)$ and new completion time $t_2 = t + Y_2$

4 Performance Metrics

Through the simulation above, the following key performance metrics are derived.

- Average Time in System (W): The average of the total time spent from entry to completing the waiting and charging.
- Average Waiting Time (W_q): The average time spent waiting in queue to use a charger.
- Charger Utilization Rate (ρ): The utilization rates for the fast and slow chargers, respectively.
- Probability of Service Denial (P_K): The proportion of vehicles that balked due to spatial constraints (N_{loss}/N_A).

5 Future Research Directions

Based on this simulation model, the following expanded scenarios can be analyzed.

- **Allowing Priority Queueing:**
  - Examine the changes in system performance metrics when priority queueing is allowed for vehicles at risk of battery discharge (under 10% remaining).

- **Applying Cooldown Times:**
  - Analyze the impact on overall waiting times when a mandatory cooldown time is applied after a certain period of continuous use to prevent charger overheating.

- **Cost-Profit Optimization Analysis:**
  - Determine purchase intent through a purchase probability function ($P(W)$) linked to charging time, calculate the expenditure

amount using a probability distribution (General RV), and then calculate the additional revenue.

- Compare the additional installation and maintenance costs of chargers with the revenue on a monthly basis to determine the optimal number of chargers and whether to expand parking spaces.